

Case Book 3

USCMarshall
School of Business

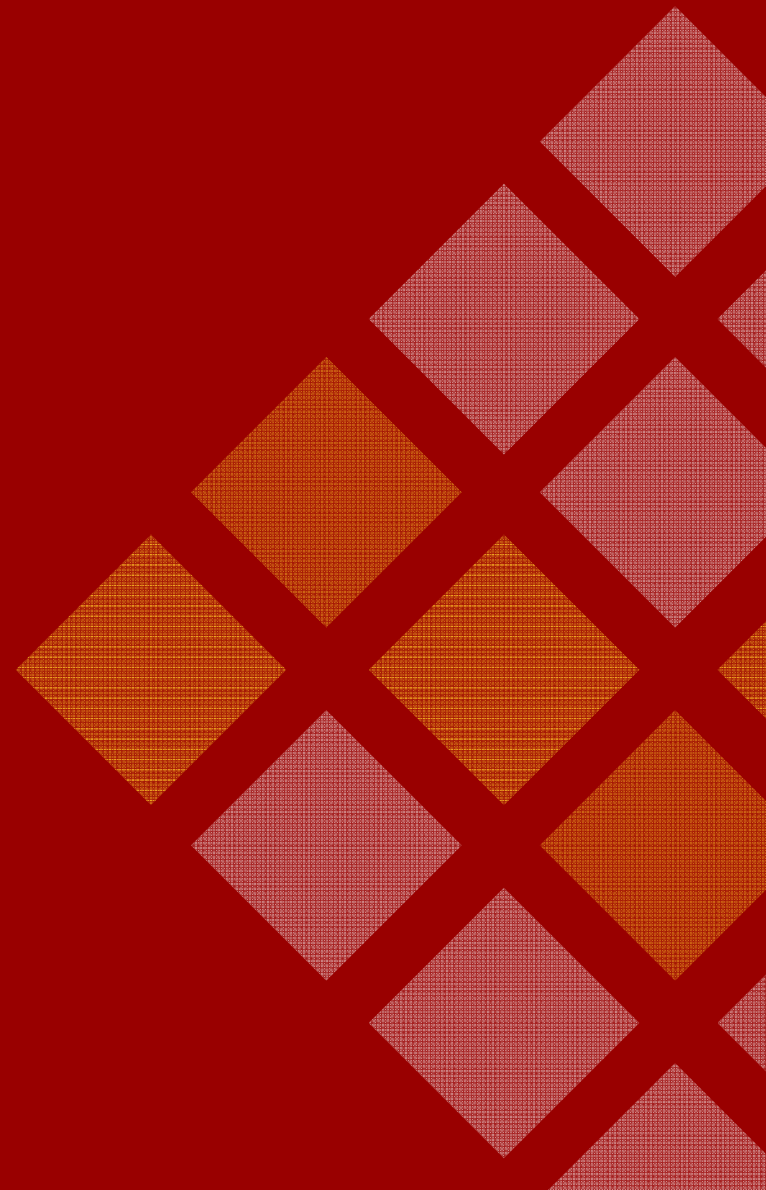


MARSHALL CONSULTING & STRATEGY CLUB

clubs.marshall.usc.edu/mcsc

Updated January 2, 2016

Case 36





Case 36: Ruby Tuesday

Prompt

Our client Ruby Tuesday, is an American fast food chain that owns, operates, and franchises Ruby Tuesday restaurants. The company was formed in 1996 as a reincorporation of Morrison Restaurants Inc. It is headquartered in Maryville, Tennessee, and operates more than 700 locations worldwide.

The CEO of Ruby Tuesday has hired us to analyze its idea of implementing a new promotion across its 500 locations in the US – a frequent diner program similar to the frequent flyer program used by airlines companies:

- For every \$1 spent in a Ruby Tuesday restaurant, the customer earns 1 point.
- For every 50 points the customer earns, he/she can redeem a free movie ticket.

Is this frequent diner program a good idea or not?



Case 36: Ruby Tuesday

Potential Structure

Additional Revenues

Gained from the implementation of the program (can come from additional customers, additional spending from a customer each time he/she visits the store, more visits per year from one customer)

Additional Costs

Necessary to implement the program (operational costs regarding the program, additional costs generated by additional meals sold, cost of the movie tickets)

Competition

How its competitors would respond to the implementation of the program by Ruby Tuesday. It seems like an easy program to implement, so it might be easy for competitors to copy.

Consumers

How consumers will respond to the new promotion, or more importantly, what their needs are, because while we are spending energy implementing this program, competitors might address other needs of the consumers that might be more likely to attract consumers to move to the competition.



Case 36: Ruby Tuesday

Additional Information (if asked)

General

- Number of Ruby Tuesday locations in U.S.: 500
- Average number of customers per restaurant: 20,000/year
- Average price of a meal: \$10
- Current profit margin: 30%
- Cost of a movie ticket: \$5 (first ask the candidate about his/her opinion on how much it would cost)
- Redemption percentage of the movie tickets: 20%
- The client would only pay for redeemed tickets.

Ruby Tuesday's Customer Segments

- Frequent customer: 40% (comes to the restaurant 30 times/year)
- Occasional customer: 60% (comes to the restaurant 9 times/year)

Program Enrollment

- 20% of the frequent customers will join the program
- 5% of the occasional customers will join the program

Increase in Number of Visits As a Result of the Proposed Frequent Diner Program

- By 6/year for frequent customers (enrolled)
- By 3/year for occasional customers (enrolled)



Case 36: Ruby Tuesday

Cost/Benefit Analysis

Benefit (additional revenue coming from the implementation of the frequent diner program)

	<u># customers</u>	<u># customers enrolling</u>	<u>Additional revenue</u>
Frequent customers	$20,000 \times 40\% = 8,000$	$8,000 \times 20\% = 1,600$	$1,600 \times 6 \text{ times/year} \times \$10 = \$96,000$
Occasional customers	$20,000 \times 60\% = 12,000$	$12,000 \times 5\% = 600$	$600 \times 3 \text{ times/year} \times \$10 = \$18,000$
Total additional revenue			\$114,000

Costs (additional costs of the extra meals sold + cost of the movie tickets redeemed)

Cost of additional revenue = $\$114,000 \times (1 - 30\% \text{ profit margin})$
= \$79,800

Cost of movie tickets = $\$5 \text{ per movie ticket} \times 20\% \text{ redemption rate} \times (1,600 \text{ enrolled frequent customers} \times 36 \text{ visits/year} \times \$10 \text{ per meal} + 600 \text{ enrolled occasional customers} \times 12 \text{ visits/year} \times \$10 \text{ per meal}) / \$50 \text{ needed to earn a movie ticket}$
= \$12,960

Total cost = \$79,800 + \$12,960 = \$92,760

Profits

Additional profits from the frequent diner program = \$114,000 – \$92,760 = \$21,240



Case 36: Ruby Tuesday

Recommendation

Recommendation

I recommend that the client should go forward with the frequent diner program as it provides the company with ~\$21,000 additional profits per store per year.

Risks

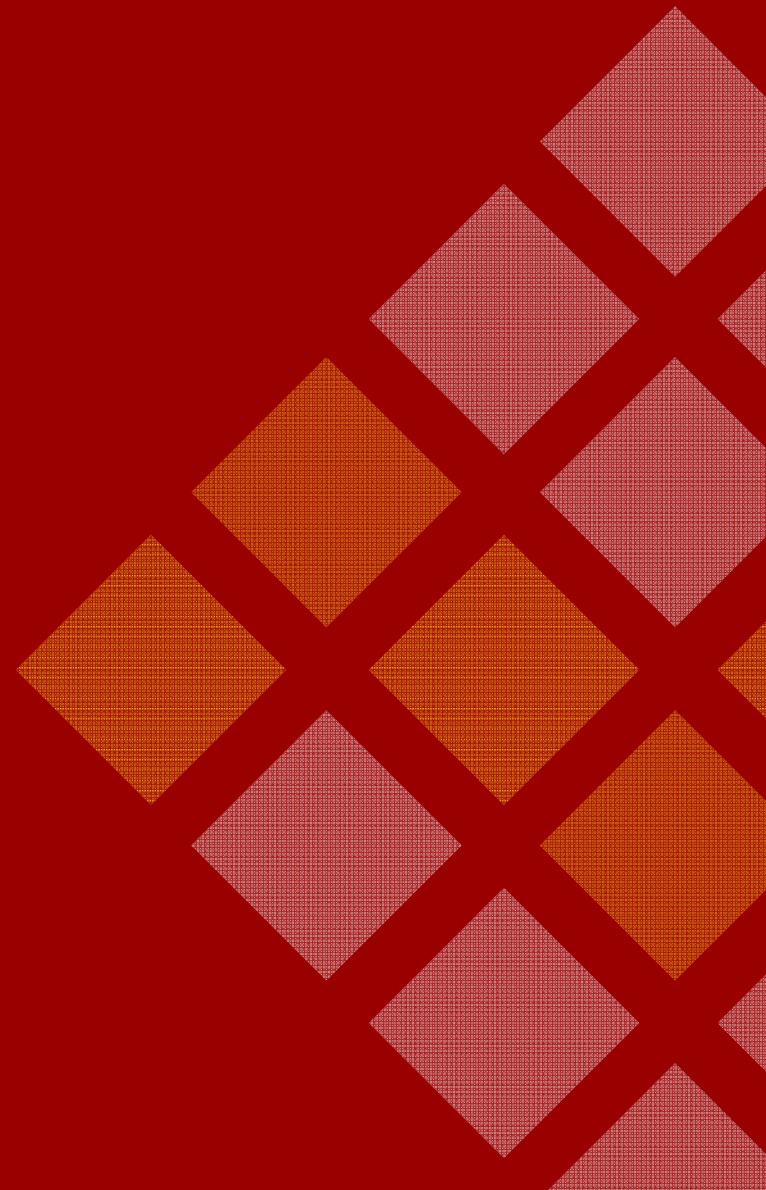
However, there are some risks involved that need to be investigated before the actual implementation, such as:

- Implementation costs (frequent diner cards, staff to help people signing up, purchasing and handling of movie tickets, filling reports, etc.)
- Longer waiting lines due to the additional tasks
- If profit margins decrease, the frequent diner program may become unprofitable (e.g. 10% profit margin)

Next Steps

As the next steps, I recommend that we investigate these aspects before going further with the implementation.

Case 37





Case 37: Chriscross Connections

Prompt

The setting is 2007. Our client is a startup with the ability to deliver broadband internet to commercial airlines. How would you help them think about their offering?

Description

This is a market entry case in which candidates are required to evaluate the feasibility of a new product in conjunction with the airline industry. The candidate should use a comprehensive framework, walk the interviewer through it, and be prepared for analytical detours throughout the flow of the case.

Calculations that are presented only represent one approach, and the interviewer should be mindful and allow for flexibility.

Additional Information (if asked)

- There is general interest in broadband internet from the airline industry.
- The startup would have to invest relatively little up front and keeps most of the revenues (client already has this technology and airlines want this service on their planes).
- The startup would charge airline passengers on a per-flight pricing model.
- The company has a patent on the high speed connection.
- Competition: There are currently no competitors in this space.



Case 37: Chriscross Connections

Analysis and Calculations

Industry and Market Size: Interviewee should mention this early on. Have the interviewee size the market for commercial airplanes. Then, inform him/her to use 3,000 planes for the purpose of this case.

After the interviewee is done sizing the market, hand him/her Exhibit 1.

Analysis of Exhibit 1

Below is one method of analyzing Exhibit 1 (there are multiple ways to analyze this).

Class	First	Coach
Seats/Plane	20	180
Load Factor	.75	.75
Full seats/plane	15	135
Biz Travelers	100%	30%
Laptop users/plane	15	40.5
Total laptop users/plane	55.5	
3000 planes x 2000 legs/plane x ~50 laptop users/plane = 300,000,000 approximate annual potential user-legs.		

Pricing: When the interviewee inquires about pricing for broadband internet on the planes, hand him/her Exhibit 2.



Case 37: Chriscross Connections

Analysis and Calculations

Analysis of Exhibit 2 (there are multiple ways to analyze this)

Taking a price*penetration ratio:

- At \$25 per flight, $25 \times 5 = 125$
- At \$20 per flight, $20 \times 10 = 200$
- At \$15 per flight, $15 \times 20 = 300$
- At \$10 per flight, $10 \times 25 = 250$
- At \$5 per flight, $5 \times 30 = 150$
- Set the price at \$15

Cost: When the interviewee asks for costs, indicate that it is \$250k per plane per year.

After the interviewee has analyzed Exhibits 1 and 2, and receives the costs, he/she should calculate profitability and observe that the start-up is profitable.

Continue pressing the candidate for other factors that he/she should consider (competitors entering market, patent expiring, new technology, government regulations, cost to increase bandwidth, etc.). Keep asking "What else?" until candidate runs out of ideas, then push him/her for a recommendation.



Case 37: Chriscross Connections

Recommendation

Recommendation

- **Expected:** Candidate performs accurate math with solid profitability and breakeven calculations.
- **Good:** Candidate achieved the "expected" category above and also provides framework that includes exploration of all possible options and offers a clear summary that summarizes recommendations, risks, and next steps in under 60 seconds.
- **Excellent:** Candidate achieved the "expected" and "good" categories above, and also considers 4-5 other factors while maintaining poise through the "what else" line of questioning at the end.



Case 37: Chriscross Connections

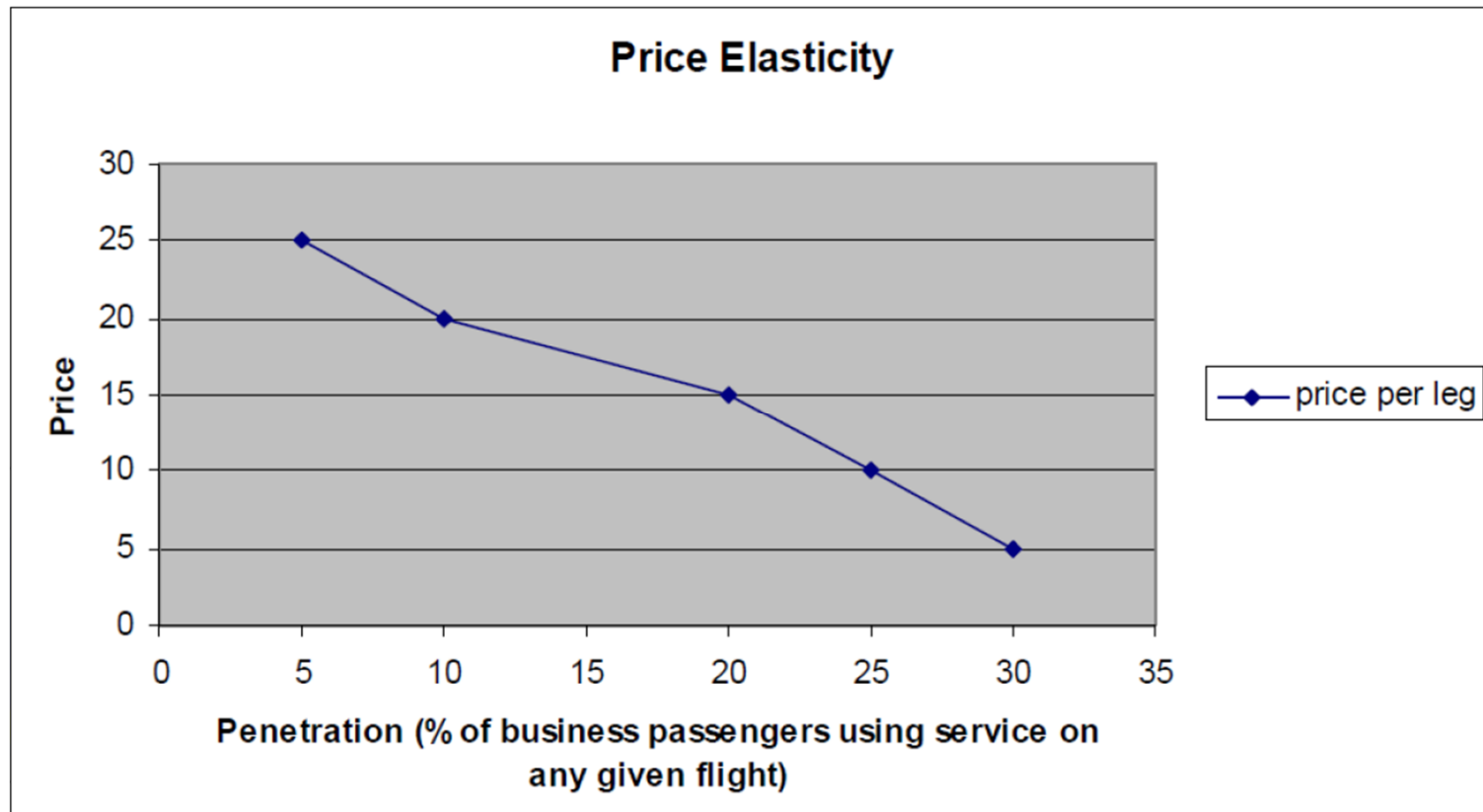
Exhibit 1

Exhibit 1			Passenger mix by cabin	
			First	
Annual flights legs per plane	2000		Business	100%
Average seats per plane	200		Leisure	0%
Average load factor	75%			
Cabin seating:			Coach	
First class	10%		Business	30%
Coach	90%		Leisure	70%
Note: 90% of business travelers carry laptops; 0% of Leisure travelers carry laptops				

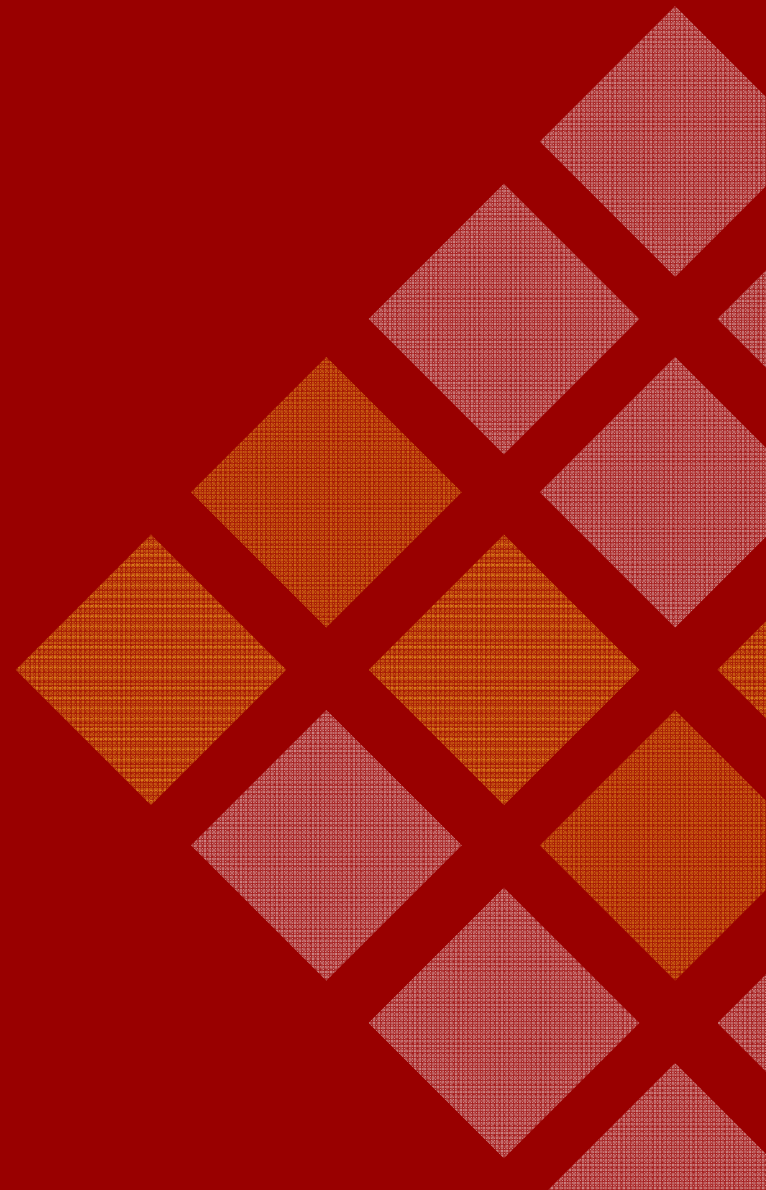


Case 37: Chriscross Connections

Exhibit 2



Case 38





Case 38: Wang's Wonderful Wings

Prompt

Our client is the owner of sports bar chain that specializes in selling chicken wings. The chain owns and operates 700 restaurants across the United States. Same store sales (SSS) declined last year, and the owner does not know why. How can our client improve SSS?

Description

This is a difficult profitability case in which candidates are required to identify the problem facing a restaurant chain. The candidate should use a comprehensive framework, walk the interviewer through it, and be prepared to analyze difficult exhibits.

The case is broken down into two halves, and it is important that the candidate cover Part 1 before Part 2. The interviewer should push the candidate to gather insights from Exhibits under 1 minute each, with the exception of Exhibit 3.

Additional Information (if asked)

- The client sells a number of food items, not just wings
- There have not been any menu changes recently
- The client has always had a strong brand image



Case 38: Wang's Wonderful Wings

Analysis

Part 1

- When the candidate asks about SSS and the main driver of that, hand him/her Exhibit 1.
 - *Do not share the following information with interviewee:* The key takeaway here is that customer traffic is highly correlated with SSS growth.
- When the candidate asks for competitor information, hand him/her Exhibit 2.
 - *Do not share the following information with interviewee:* The key takeaway here is that there is a limited opportunity to increase price or bill size.
- When the candidate asks about operational inefficiencies or table configurations, hand him/her Exhibit 3.
 - NOTE: Do not show Exhibit 3 until candidate alludes to some sort of inefficiency.
 - *Do not share the following information with interviewee:* The key takeaway is that there is an opportunity here to reconfigure the restaurants to accommodate more parties of one or two.
 - Offer very little assistance (preferably none) with Exhibit 3. The graph is intentionally difficult to interpret.
- After candidate is done with Exhibit 3, ask him/her to calculate the utilization of the tables during peak hours. The answer should be ~50%.
 - *Do not share the following information with interviewee:* The data shows that 4-seater tables are ~50% utilized during peak hours. About $\frac{1}{4}$ of the 4-seat tables are occupied by 3 people, and about $\frac{3}{4}$ of the 4-seat tables are occupied by 2 people. Thus, $\frac{1}{4} * (75\% \text{ occupancy}) + \frac{3}{4} * (50\% \text{ occupancy}) = \sim 50\% \text{ utilization}$. The specific number would be 56.25%.



Case 38: Wang's Wonderful Wings

Analysis

Part 2

- Probe the candidate for multiple ways to increase utilization. Keep probing until candidate has named 5 ways or runs out of ideas.
- Afterwards, ask the candidate the following, "What is the bottom line impact of separating one 4-seater into two 2-seaters in every restaurant?"
- Analysis:
 - Additional seats: 2
 - *Let candidate make this assumption:* Peak hours: 2
 - *Let candidate make this assumption:* Table turns per hour: 2
 - Additional meals per restaurant: $2 \times 2 \times 2 = 8$
 - 8 meals/restaurant x 700 restaurants = 5,600 meals
 - 5,600 meals x \$50/average meal = \$280,000 revenue
 - \$280,000 revenue x 30% margin = \$84,000 additional profit/day
 - \$84,000/day x 350 days/year = \$29.4M additional profit/year



Case 38: Wang's Wonderful Wings

Recommendation

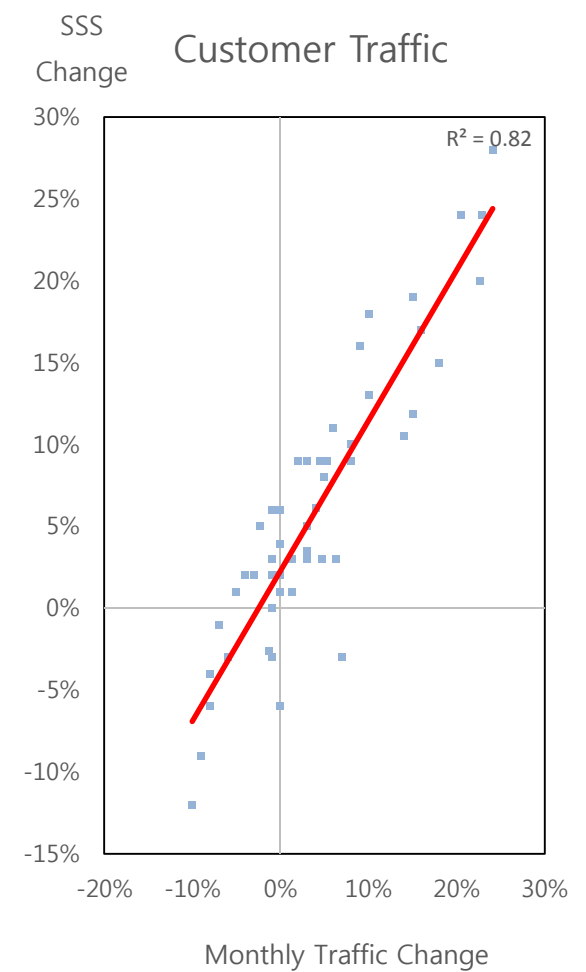
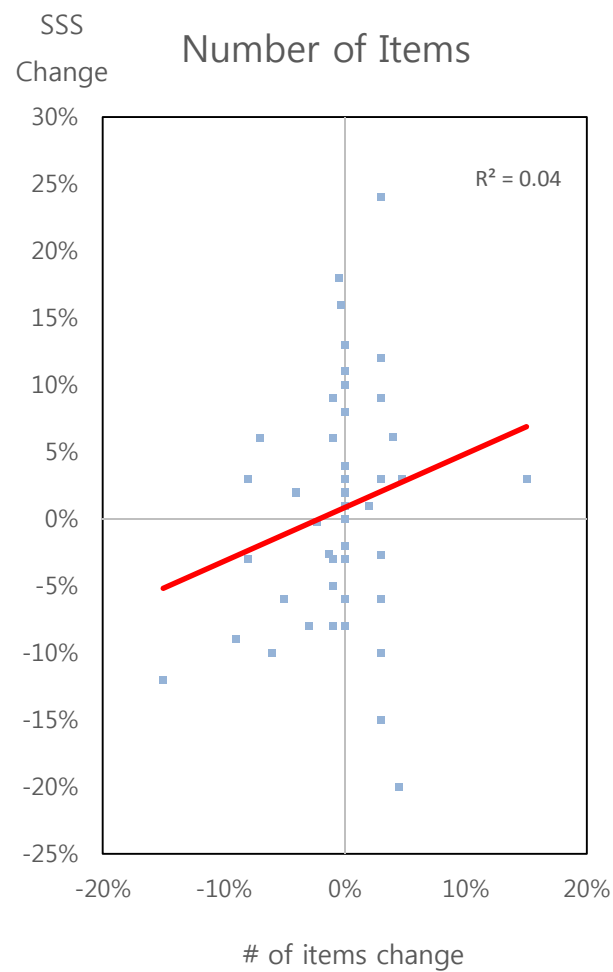
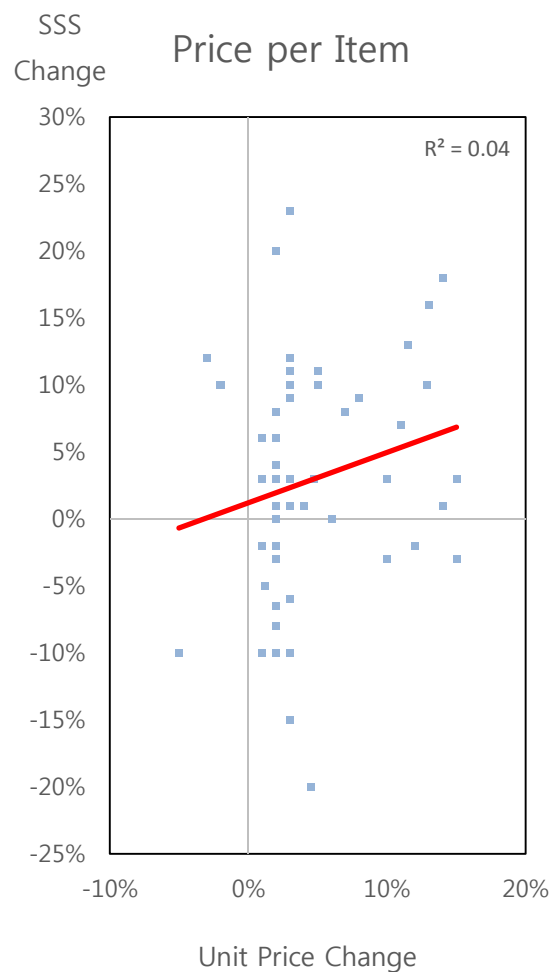
Recommendation

- **Expected:** The candidate will interpret exhibits and offer insights while viewing them. The candidate will also recognize the opportunity to optimize table configurations. Candidate should also maintain composure throughout the entire case as they are probed for ideas.
- **Excellent:** Candidate will do all of the above but also probe about table configurations without assistance. Candidate should also calculate bottom-line impact without assistance. Candidate should also offer a clear, crisp recommendation that lasts for no longer than 45 seconds.



Case 38: Wang's Wonderful Wings

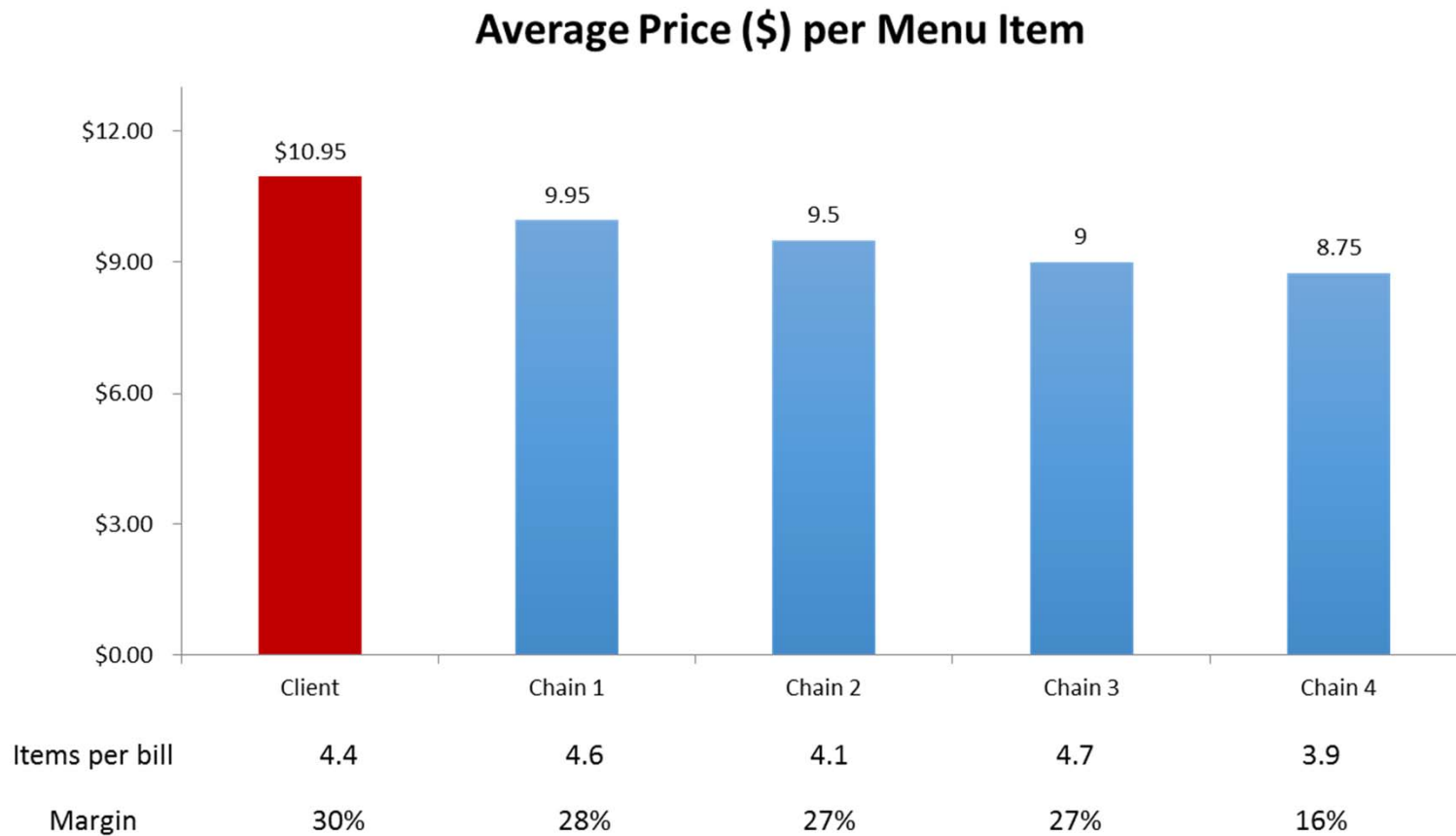
Exhibit 1





Case 38: Wang's Wonderful Wings

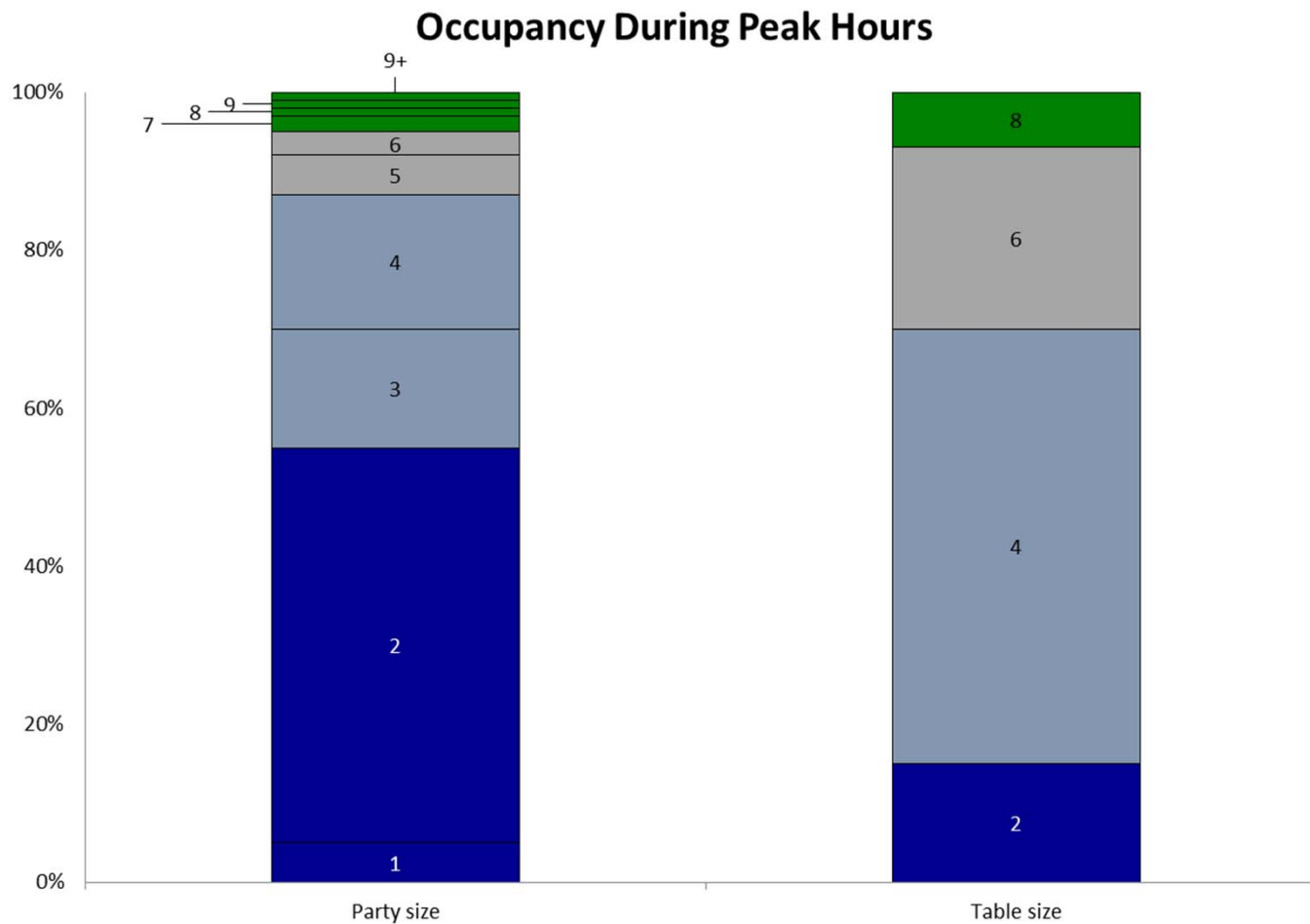
Exhibit 2





Case 38: Wang's Wonderful Wings

Exhibit 3

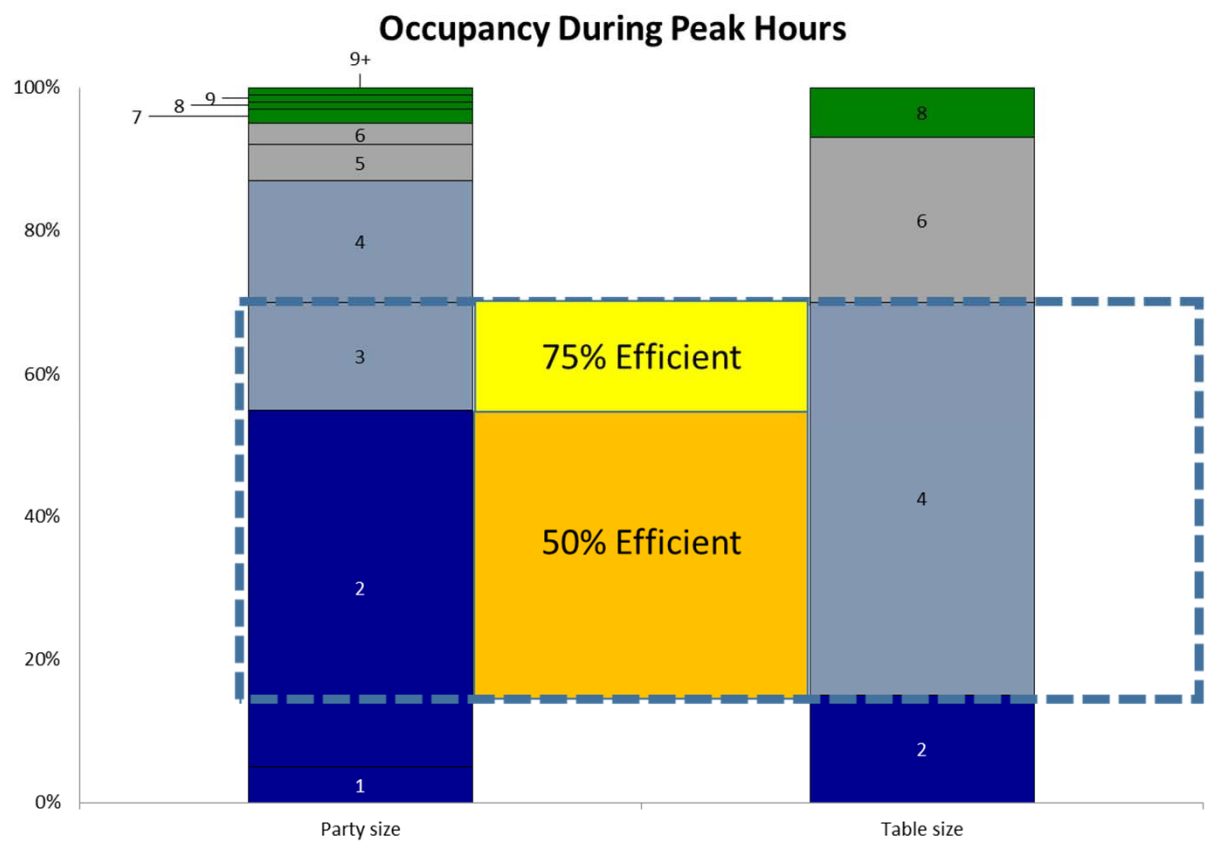




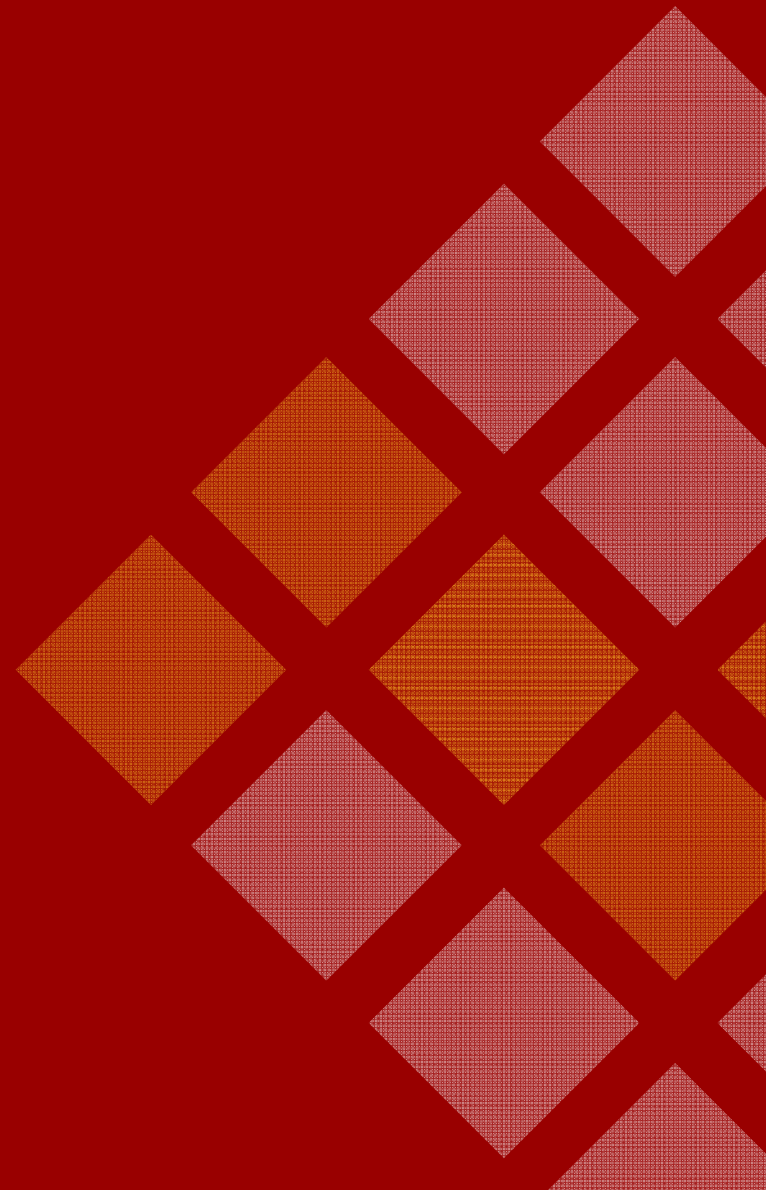
Case 38: Wang's Wonderful Wings

Exhibit 3 – Interviewer Copy

INTERVIEWER COPY – DO NOT GIVE CANDIDATE



Case 39





Case 39: Digital Ice Cream

Prompt

Our client is a fresh MBA just graduated from business school. Encouraged by the recent technology and Internet startups boom, and also to make the world a better place, the client has just founded a startup company together with two business school classmates. Their business plan is very simple: to start an online ice cream delivery shop.

Similar to other online food ordering companies such as GrubHub, Seamless, Delivery Hero, Just-Eat, FoodPanda, etc., the client's startup works as follows: They would buy an ice cream store first, then build a website that would list both the kinds of ice cream that they sell and menus from local ice cream shops. Users would use the website to place orders online. The client would then be responsible for arranging the ice cream delivery, either from the client's own store or from local ice cream shops. The client would generate revenues by charging a small commission fee to ice cream shops for every customer order from their website.

Is this a good idea? Why or why not?



Case 39: Digital Ice Cream

Sample Framework / Factors

1. Internal Factors / Value Chain
 - a. Supplier Types
 - Local ice cream shops
 - Third party bulk ice cream distributors (wholesalers)
 - Buy ice cream at store and store in freezer for internally controlled distribution
 - b. Distribution
 - Rely on local shops (similar to online florists)
 - In company owned freezer trucks
2. External factors/ Market
 - a. Advertising
 - Where to advertise? Bonus for suggesting real potential options such as Google AdSense
 - Which markets and segments to target?
 - b. Delivery Methods
 - Rely on local ice cream shops {similar to online florists}
 - In company-owned freezer trucks
 - c. Competitors
 - Market Size (will people order ice cream online, and why?)
 - Grocery stores and other brick-and-mortar ice cream shops
 - d. Seasonality
 - Can you achieve sales year-round or only during hotter months
 - Changing ice cream flavors, and therefore inventory, by time of year



Case 39: Digital Ice Cream

Analysis

Part 1: Ask the candidate to estimate the market size (in \$) in the United States for online ice cream delivery (averaged over the year).



Case 39: Digital Ice Cream

Analysis (continued)

Suggested Solution

1. Determine the population of the country (if asked: approximately 300 million)
2. Calculate the portion of population that eats ice cream
 - There are many types of ice cream and it is a popular food item
 - Roughly half of the population eats ice cream regularly, and it is evenly distributed across income, age, and gender
 - $50\% * 300M = 150 \text{ million}$
3. Find the part of population that would be willing to order ice cream online
 - 20-30 year old population is more likely to order online, especially for gifts
 - Average life expectancy is around 80 years, so percentage of population in the target age range would be $(30-20)/80 = 12.5\%$
 - $12.5\% * 150M = 18.75 \text{ million}$
 - Percentage of this age group population that would order online is 20%
 - $20\% * 18.75M = 3.75 \text{ million}$
 - To simplify calculation, let's say 4 million
4. Estimate the average size of the order (price per order) to find the total market size (note: exact numbers are not important, provide them if asked)
 - Customers are likely to order larger quantities if ordering online due to delivery costs
 - Average price of ice cream is around \$3 per pint (to be provided if asked)
 - Average online order is 2 pints, giving an average order of \$6
 - Average person orders once or twice a year
 - $4 \text{ million} * \$6 * 2 = \48 million



Case 39: Digital Ice Cream

Analysis (continued)

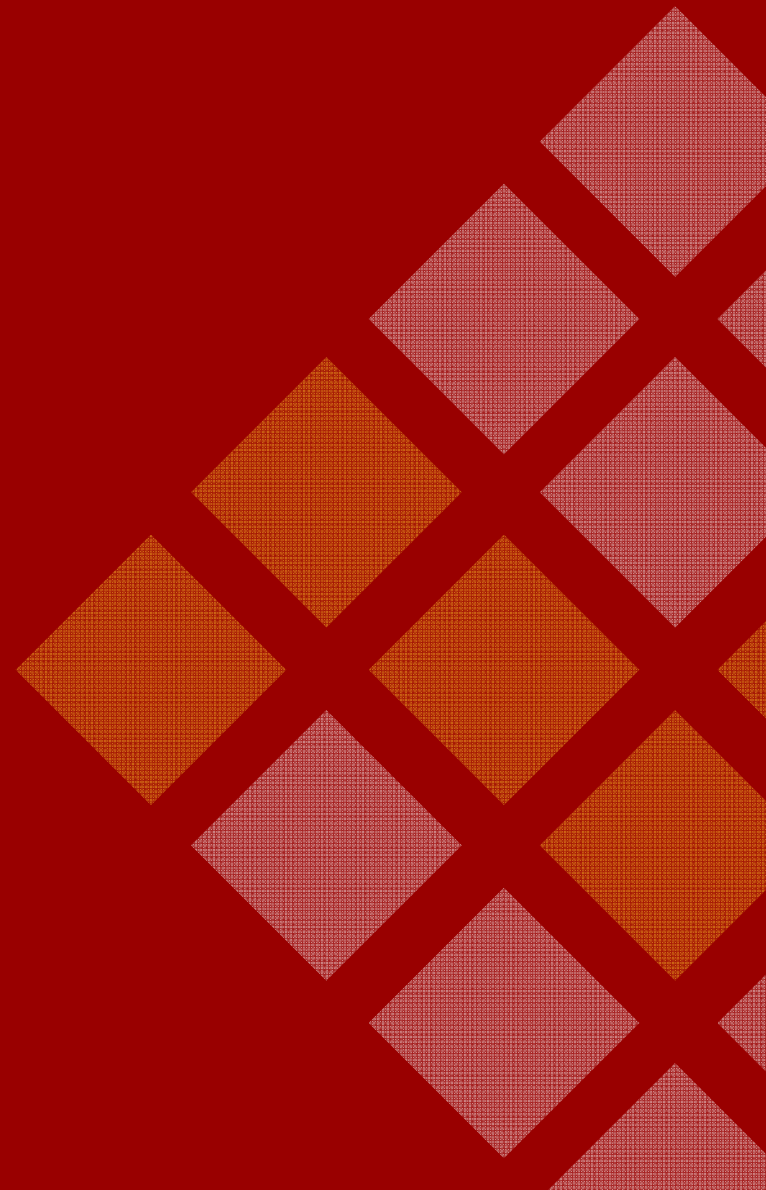
Part 2: Ask the candidate if this is an attractive business.

Suggested Solution (more detailed answer preferred):

The short answer is no. Main reasons are:

- Market size is too small: only \$48 million in total addressable market
- Business model still unproved: will people really order ice cream online?
- Competition from other brick-and-mortar ice cream shops.

Case 40





Case 40: Tech Takeover

Prompt

Yanda is a freemium enterprise social networking service that was launched in 2008. Yanda is used for private communication within organizations and is an example of enterprise social software. The tool was originally developed as an internal communication system for the genealogy website Findo. Access to a Yanda network is determined by a user's Internet domain so that only individuals with appropriate email addresses may join their respective networks.

The year is 2012. Technology giant Microsoft is looking to buy out Yanda. At this time, Yanda is already profitable and is expected to have steady, continued growth for the foreseeable future. The CEO of Yanda has offered to sell for \$1.2 Billion. Is this a reasonable price for Microsoft?

Description

This should be a fairly quick case to complete. The candidate should outline a framework that breaks down things to consider when valuing an M&A scenario. Push him/her to tell you all possible ways to value a firm if not included in framework then ask him/her to perform a simple DCF calculation (only give perpetuity formula after letting him/her try to figure it out).

Ask the candidate to brainstorm other considerations and continue to ask "What else?" until out of ideas before asking him/her to wrap up the final recommendation.



Case 40: Tech Takeover

Analysis Part 1

What are some methods for valuing the target?

Suggested Solution:

- Discounted Cash Flow (DCF)
- Comparables / multiples from similar companies.
- Multiples such as price to earnings ratio, price to sales, assuming that information is available.

Analysis Part 2

Using a simple DCF, is \$1.2B a good deal?

Simplified Discounted Cash Flow = Perpetuity = Cash Flow / (Discount Rate – Growth) = $\$50 / (0.11 - 0.03) = \625 million

Therefore, the asking price of \$1.2 billion is not a good deal.

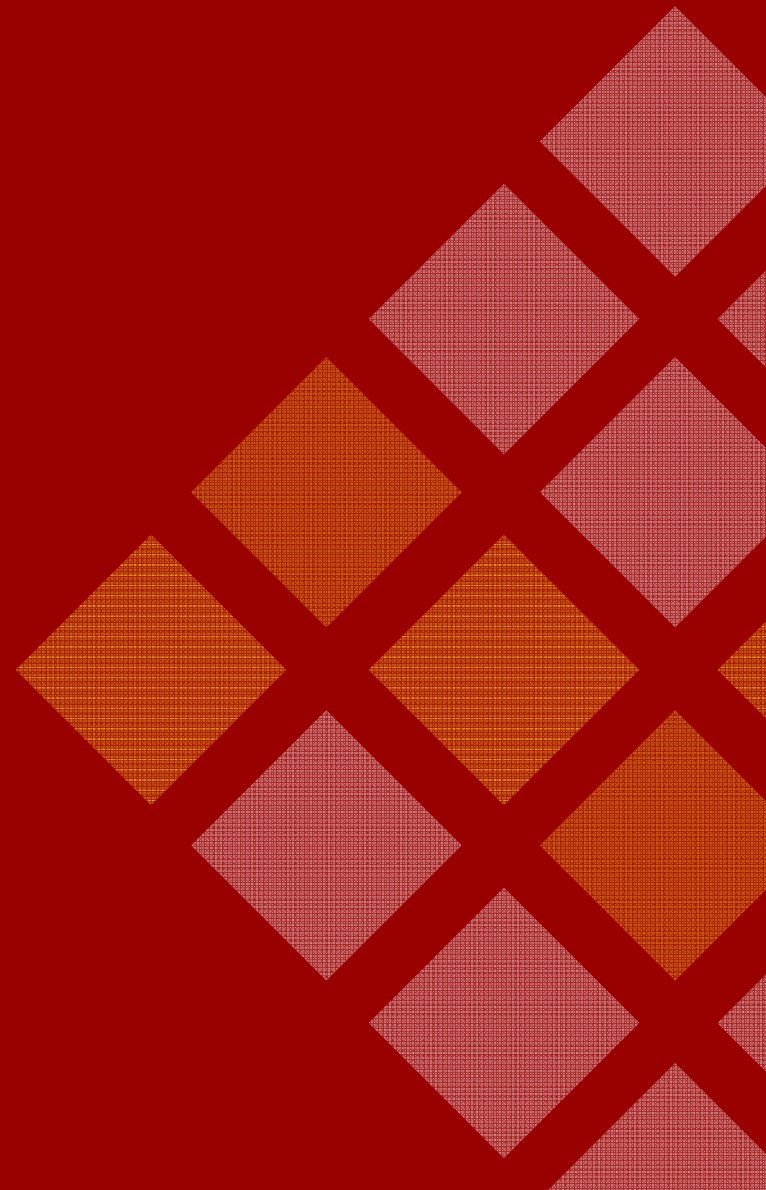
Analysis Part 3

What else should be considered?

The discount rate of 11% is quite high, is it possible that Yanda is less risky and deserves a lower discount rate?

Can Microsoft add any value to Yanda that will increase its future cash flow, lower costs at Microsoft, or add other synergies that might increase the value of the acquisition?

Case 41





Case 41: Galileo Telescopes

Prompt

You're a consultant in 17th century Venice, Italy and PowerPoint is your art canvas. You have just been hired by Galileo Galilei (15 February 1564 – 8 January 1642). Galileo was an Italian physicist, mathematician, astronomer, and philosopher who played a major role in the scientific revolution. One of his greatest achievements include improvements to the telescope and consequent astronomical observations and support for Copernicanism.

The year is 1609. Your client Galileo Galilei has just invented a new and more powerful telescope, with up to about 30x magnification. He'd like you to help him make some money off what he is sure will be a very useful invention. What do you tell him? How much money would Galileo make from telescope?

Additional Information

Galileo has a workshop, and 6 talented workers who can manufacture the telescopes for him. Each worker is available 6 days a week, for 8 hours a day, and all-in labor costs are \$10 per worker per hour. Raw materials for a telescope come to \$5 per piece, and one worker can make one telescope in 2 hours. This data is only for the costs section of the problem.

Skills involved in making a telescope consist of grinding the lenses correctly, building the housing and focal length adjustment mechanism, and assembling the parts.

The market being considered is the entire world.

No data is available on the populations of Italy, Venice, military, traders, frequency of travel, building, warfare, etc. The interviewer can respond to all requests for specific information on this stuff with "Why don't you try and derive an estimate for that?" As long as the assumptions are reasonable, most answers are valid.

This is a brainstorming case – the interviewer is looking for creativity in an unusual context. Cases are not dependent on historical facts so any inaccuracies in the candidate's display of 17th century Europe knowledge is given leeway.



Case 41: Galileo Telescopes

Potential Framework & Thoughts for Each Area

1. Company

- Part of the "Company" bucket includes costs.
- Work out costs and production capacity based on the data above.
- Could suggest cost-based pricing
- The candidate would also need to discuss the distribution strategy, which would likely involve selling the goods to local merchants for overseas trading. Also, the company cannot likely produce to keep up with initial demand so the good should be priced accordingly. Should more laborers be hired to increase production?

2. Customers: Who would need a telescope?

I. Local

- Traders and sailors
- The military – army/navy
- Explorers – land/sea
- Architects and surveyors

II. Overseas

- Geographic categories
- Europe is viable
- Asia, Middle East, America, Far East secondary markets
- Could rank in an Effort/Return quadrant matrix (some firms loves their matrices)
- Political considerations
- Profit sharing with traders
- Overseas monopoly
- Large potential demand, relatively slow supply
- High likelihood of local imitators

III. What do you think is the largest market, at each stage of the product timeline?

- Extra credit: Very rough market-sizing (not the focus of the case)
- Could suggest (very rough) pricing based on willingness-to-pay

3. Competitors

- No patent structure means imitation is a serious problem.
- Ease of construction means reverse-engineering is not difficult. Obfuscation of the mechanism may be necessary.
- 6 workers are skilled engineers. Each is an intellectual property (IP) risk if they break away and start up on their own – keep them happy and well compensated.



Case 41: Galileo Telescopes

Cost Analysis

The math is incidental, and not the focus of the case. However, the interviewer should test the candidate's quantitative ability using the example below. The candidate should be able to express why he/she would want to calculate the cost. Reasons can include the desire to assess how many telescopes can be produced relative to an estimated market demand or estimated financial resources required to operate the business.

Interviewer has already provided the following data:

- Six talented workers who can manufacture the telescopes.
- Each worker is available 6 days a week, for 8 hours a day
- All-in labor costs are \$10 per worker per hour.
- Raw materials for a telescope come to \$5 per piece
- One worker can make one telescope in 2 hours

Expected calculations:

- Cost per unit telescope = $\$5 + 2 * \$10 = \$25$
- One week: $6 \text{ workers} * 8 \text{ hour/day} * 6 \text{ days/week} = 288 \text{ worker-hours}$
- # of telescopes manufactured in that time = $288 / 2 = 144 \text{ telescopes}$
- Total raw materials cost = $144 * \$5 = \720
- Total labor cost = $288 * \$10 = \2880
- Total weekly cost = \$3600 for 144 telescopes



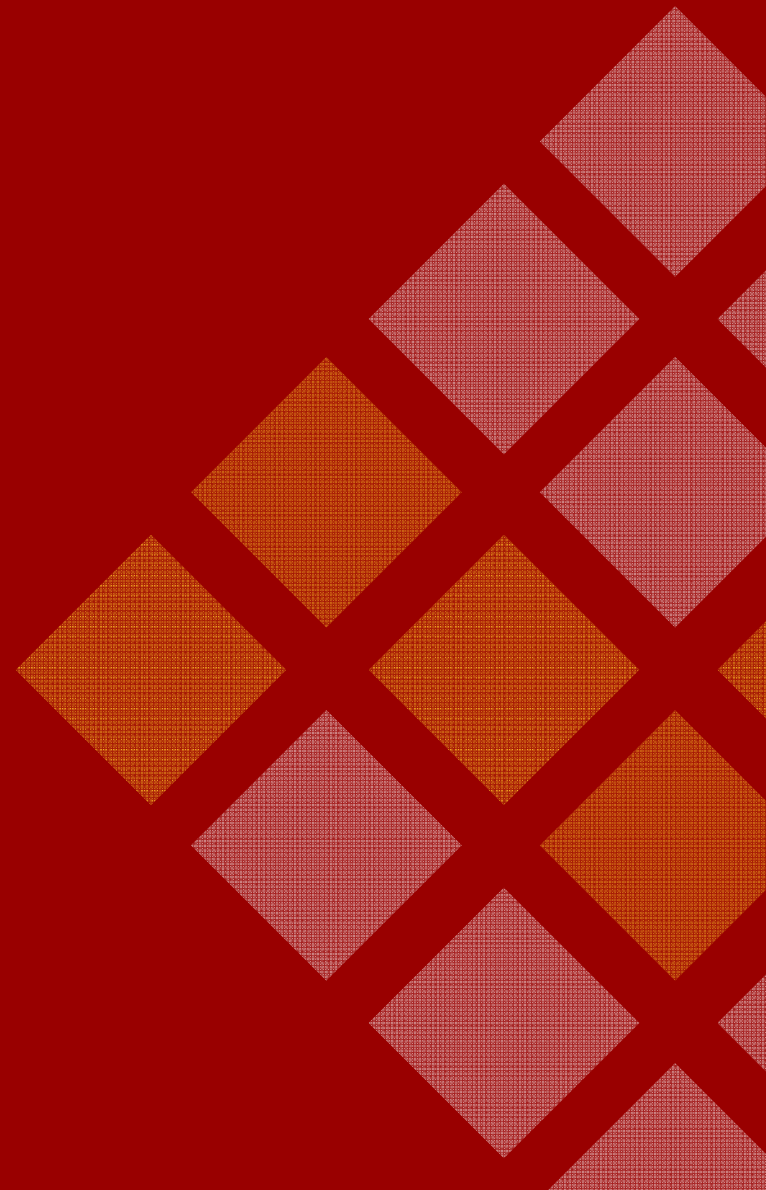
Case 41: Galileo Telescopes

Recommendation

A large market exists, both locally and overseas. The market will experience a short initial period of monopoly after which imitators will provide serious competition – preventive strategies are necessary to combat this.

Structure the response, but be as creative as possible in terms of your ideas. The math is incidental, and not the focus of the case.

Case 42





Case 42: Chicago Hotspot

Prompt

Your client is an entrepreneur who is looking to invest in new bar in downtown Chicago. He needs your help to determine how profitable the bar would be, and convince his primary investor that it would be a viable business. What are the areas you would consider/investigate for a sports bar in Chicago? What recommendation would you give to your client?

Additional Information

1. Investment standpoint
 - Is there an expected return? What return do you think you can make?
 - Is your client's father going to be active in management? No.
2. Customers/Marketing
 - What type of bar is it going to be? Sports bar.
 - Who is my customer base? Upscale men, aged 25-40.
 - Will it be open at night, day, with food/no food? Sunday – Wednesday (noon-12am) Thurs – Sat (noon – 2am), will have food service.
 - What is the product mix? Food and drink.
3. Location: Chicago (Interviewer leads: How would you find a good place in downtown Chicago? What resources would you use?)
 - Proximity of competitors
 - Talk to real estate agents, figure out what type of properties are available
 - Where does customer segment live
 - Segment by district



Case 42: Chicago Hotspot

Analysis

Expected Calculations on the Viability of New Sports Bar

1. Revenue

- a. Food – Interviewer asks how would we project how much we might earn
 - i. Look at capacity
 - ii. Benchmarking
 - iii. Average amount spent on food (\$15)
 - iv. 10 Food customers / hour
- b. Drinks
 - i. How much do they spend on drinks on average per order? (\$20/ each)
 - ii. 4 drinks per customer / hour

2. Costs

- a. Buy Rent or Lease (Lease \$10,000 / month)
- b. Labor (3 types of people)
 - i. Kitchen - \$10 / hour, 4 people when kitchen opens
 - ii. Bar – Need one bartender at all times, because bartender works on tips they get \$5 / hour
 - iii. Wait Staff – 3 at all times at \$5 / hour
- c. Utilities (\$200 / month)
- d. COGS (20% of sell price)
- e. Other (Legal, insurance, licenses, training, equipment) (\$1,000 / month)

3. Profits

- a. Revenues – Costs = \$21,000 - \$20,720 = \$260 / month = \$3,120 / year



Case 42: Chicago Hotspot

Recommendation

Recommendation

Encourage interviewee to talk through if they think that monthly profit would be a good investment when considering start up costs. No start up costs are given so there is some leeway into what they can recommend.

A recommendation should include either a go / no go.

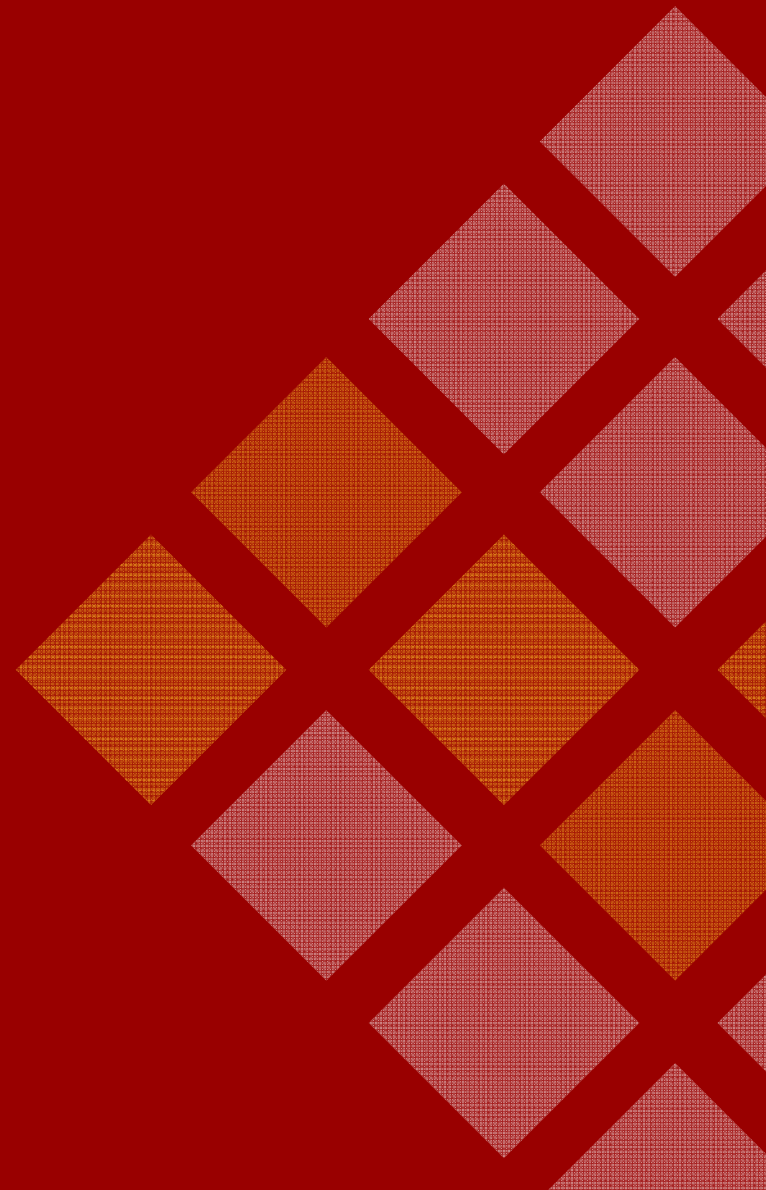
Risks

Risks include an uncertain start up cost and profit estimations.

Next Steps

Next steps should mention how to best locate the best neighborhood to bring in the restaurant and how to best ensure maximum sales.

Case 43





Case 43: GM Goes to Brazil

Prompt

The General Motors Financial Company is a financial services arm of automaker General Motors. The company is a global provider of auto finance, with operations in the US, Canada, Europe and Latin America. The company is headquartered in Fort Worth, Texas. GM Financial provides retail loan and lease programs through auto dealers for customers across the credit spectrum. They also offer commercial lending products, such as retail floorplan, construction and real estate loans, or insurance for car dealerships.

For this case, your client is GM Financial. General Motors recently entered the Latin American market and GM Financial has started to finance vehicles to final consumers in Brazil. The CEO of GM Financial wants to structure a new call center for these new clients in Brazil and has hired your team to help.

Description

This case has both market sizing and market entry parts.

This case is also a heavy quantitative case.



Case 43: GM Goes to Brazil

Additional Information (if asked)

- The call center handles just inbound calls; it receives calls of current customers only. There is no need to consider prospective clients.
- As other channels can be used to access the service, assume that only 40% of the customer base in Brazil would utilize the call center. Consider also just one call per customer per year.
- The parent company General Motors has 20% of the market share in Brazil.
- On average, 80% of new cars are financed.
- 50% of financed cars use the financial services arm of the car manufacturer.
- On average, the financial service is provided for 48 months.
- On average, a customer buys a new car every 4 years.
- On average, a phone call lasts for 5 minutes.
- Each call center attendant works 6 hours a day.
- The call center operates 5 days/week, from 6 a.m. to 8 p.m.
- Consider at least 2 shifts per day.
- Average household in Brazil = 4 people/family
- Average # of car/family = 1
- On average, a family changes its car every 4 years (new vehicles).



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Analysis

Question 1: Estimate how many attendants are needed in the call center to serve customers in Brazil.

1. Market Size

- Population of Brazil = 200M
- # of household = $200M / 4 = 50M$
- Assume that high-income families = 10% and medium-income families = 50%
- # of household that owns car = $50M * 60\% = 30M$
- Assume that 60% of them buy new car; 40% buy used car
- # of years on average that a family waits to purchase a new car = 4
- New cars sold every year = $30M * 60\% / 4 = 4.5M$

2. Customer Base

- GM's market share in Brazil = 20%
- GM new cars sold per year = $4.5M * 20\% = 900K$
- Financed cars (%) = 80%
- Financed cars (units) = $900K * 80\% = 720K$
- Cars financed by GM Financial (%) = 50%
- Cars financed by GM Financial (units) = $720K * 50\% = 360K$
- Customers base (each car is financed for 4 years) = $360K * 4 = 1.44K$
- Only 40% of the customer base makes call = $1.44K * 40\% = 576K$

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Analysis (continued)

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3. Call Center Capacity

- Average minutes/call = 5
- Hours worked/attendant/day = 6
- Days worked/attendant/week = 5
- Calls/week/attendant = 5 (days/week) * 6 (hours/day) * 60 (min/hour) / 5 (min/call) = 360
- Calls/year/attendant = 360 * 52 = 18,720

4. Attendants needed = $576,000 / 18,720 = \sim 31$



Case 43: GM Goes to Brazil

Analysis (continued)

Question 2: GM Financial now wants to estimate operations of the call center if it were to be open 24 hours/day, 7 days/week. How many employees should be hired (employees now working 7 days a week)?

Assuming fixed demand of 576K calls per year:

- Total demand/year (min) = $576K * 5 \text{ min/call} = 2.88M$
- New days worked/attendant/week = 7
- Total supply/attendant/year (min) = $52 \text{ week} * 7 \text{ (days/week)} * 6 \text{ (hour/day)} * 60 \text{ (min/hour)} = 131,040$
- Attendants needed = $2.88M / 131,040 = \sim 22$

Therefore, assuming fixed demand, the call center can work with only 22 employees. However, this assumes that every call center attendant works 7 days/week with no vacation. An excellent candidate should demonstrate that he/she understands the different processes and constraints in a call center, and how these affect overall productivity, rather than just calculating numbers.



Case 43: GM Goes to Brazil

Recommendation

Assuming there are some peak hours, the candidate can increase the number of attendants to 25 or 26. The candidate must comment about the variation of inbound call distribution: working days vs. weekends/holidays/sleep time, etc.

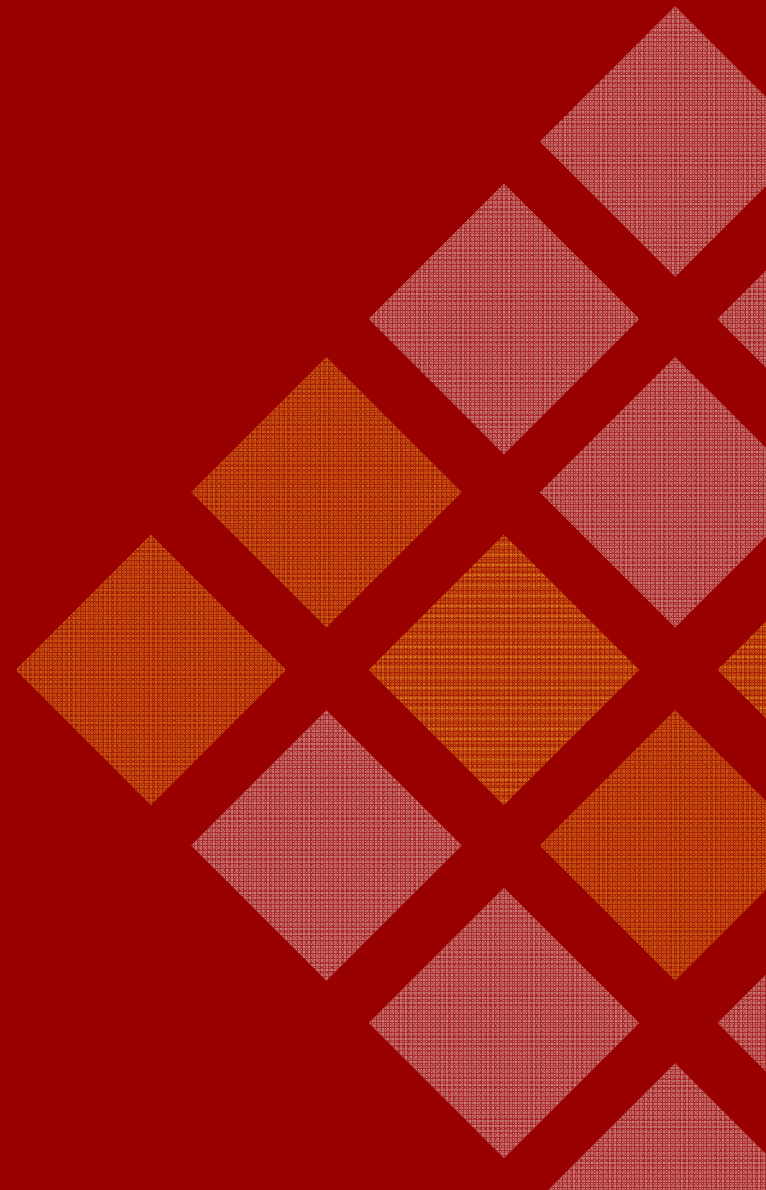
The candidate may propose intermediate solutions (7 days/week, but not 24 hours/day), since late night and early morning hours (i.e., 10PM – 7AM) should have almost no inbound calls.

Inbound calls to the call center should include both current customers and potential customers (who call to ask for information but don't ultimately become customers). Potential customers are not included in the above calculation.

As time/call is assumed to be 5 minutes, no unproductive or administrative time is considered and therefore might be a concern raised by the candidate.

After completing a call with a customer, some additional work may be required for the call center attendant, such as entering data to a system, etc. This, also, has not been considered in the above calculation.

Case 44





Case 44: Internet Routers in the US

Prompt

A network router is a device that forwards data packets between computer networks, creating an overlay internetwork. The most familiar type of routers are home and small office routers that simply pass data, such as web pages, email, messages, and videos between home computers and the Internet. An example of a router would be computer owner's cable or DSL modem, which connects to the Internet through an Internet service provider (ISP).

Estimate the annual market size for internet routers in the commercial segment (companies with 20 to 1,000 employees) in the U.S.

Additional Information (if asked)

For companies with between 20 to 1000 employees, IT spending is about 5% of a company's annual revenue.

Networking is a subset of IT, and networking spending is about 20% of total IT spending.

Internet routers make up 5-10% of spending on networking.



Case 44: Internet Routers in the US

Analysis

This "market sizing" case can be solved by starting from the U.S. population. First, let's estimate the number of U.S. companies with between 20 and 1000 employees:

- U.S. population: 300M.
- Assume half are of working age, so 150M potential employees in the U.S.
- Of those 150M people, assume 2/3 are employed full-time, or 100M.
- Assume 60% of full time employees fall into the segment we're looking at, i.e. working for companies with between 20 to 1,000 people. This gives us 60M.
- Assume an average of 500 employees per company, so $60M / 500 = 120K$ companies.

Then, make an estimate for revenue per head at these companies.

- Let's say, on average each employee makes \$80K contribution to company's revenues.
- So, at a given company, annual revenue per company is $\$80K * 500 \text{ employee} = \$40M$.

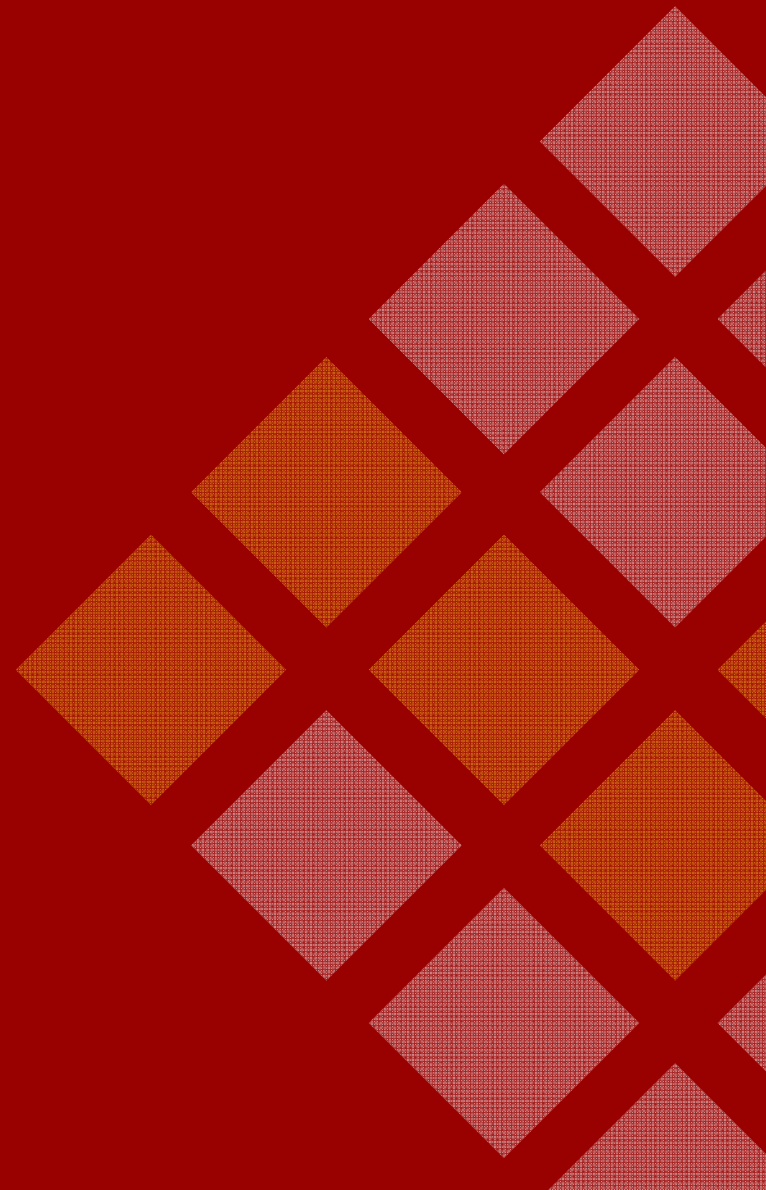
Next, use the additional information provided to calculate the share of revenue spent on internet routers by a given company.

- $5\% * 20\% * 10\% = 0.1\%$ of total annual revenue.
- So for each company, annual spending on routers is $\$40M * 0.001 = \$40K$.

And for the total market, we just multiply that by the number of companies, already estimated to be 120K.

- $120K * \$40K = \$4.8B$ estimated market size for routers in U.S. commercial segment.

Case 45





Case 45: IBM Meets Lenovo

Prompt

The year is 2004, and you have been hired by the International Business Machines Corporation (IBM) to support them with their portfolio strategy. Headquartered in Armonk, New York, IBM manufactures and markets computer hardware (such as the IBM Lenovo Thinkpad) and software, and offers infrastructure, hosting, and consulting services in areas ranging from mainframe computers to nanotechnology.

Currently (in year 2004) IBM has 10 lines of business, including personal computers, servers, mainframe computers, software, networking equipment, IT services, consulting services, etc. We are evaluating to understand which businesses our client should divest from, and what should be the strategy for the ones they keep. Specifically, our team is responsible for analyzing IBM's personal computers (PC) division. Should our client IBM invest in or divest its PC division?



Case 45: IBM Meets Lenovo

Additional Information

1. PC Market Environment (in 2004)

- Market growth trends
 - Average 4% per year sales growth
 - 15% per year for notebook/laptops
 - -5% per year for desktop PCs
- Fragmented Industry: 10+ players (Dell, HP, IBM, Acer, Fujitsu, Toshiba, Lenovo, NEC, Apple, Sony, etc.)
- Market Shares and Growth Trends
 - Competitor A (Dell): ~20% and slowly growing
 - Competitor B (HP): ~13% and slowly growing
 - Our client (IBM): 5% and slowly declining
 - Other competitors: ~60%

2. Client Company

- IBM has a very strong brand that is synonymous with quality and state-of-the-art technology, but so far, its brand has not translated into an ability to charge a price premium on its products.
- Manufacturing is currently 95% based on pre-configured systems.
 - IBM has not developed the capability to deliver customized systems in an adequate timeframe for end users.
 - IBM sells PCs through its website, but online sales are negligible.
- IBM has its own sales force to cater to corporate customers, but penetration for corporate customers is very low and this segment is not profitable for our client at this point.
- The Personal Computers (PC) division has no specific strategic importance to IBM.



Case 45: IBM Meets Lenovo

Additional Information

3. Competitors

A. Competitor A (Dell)

- Competitor A's (Dell) core business is personal computers.
- Dell uses a direct model for consumers and SME, and has a large and respected sales force for corporate clients.
- Dell focuses on a made-to-order model, having inventory turns that are 3X that of our client, IBM, and 2X that of Competitor B, HP, and is renowned for its operational effectiveness.
- Due to bargaining power with suppliers and its made-to-order model, Dell has negative working capital.
- Despite its lower gross margin percentages, Dell still has EBITDA margins that are 2X the industry average (where the industry average includes Dell).
- The industry's average EBITDA margins, not including Dell, is negative.

B. Competitor B (HP)

- Competitor B's (HP) core business includes personal computers, servers, networking equipment, and IT services.
- HP sells through big box retailers to consumers, and through its own sales force to corporate clients.
- HP sells mostly pre-configured systems, but also delivers made-to-order computers.
- HP has above-market-average EBITDA margins and inventory turns.



Case 45: IBM Meets Lenovo

Additional Information

4. Customers

	Consumer (30% of the laptops market)	Corporate (70% of the laptops market)
Preferred Channel	Direct service or retail	Specialized sales force
Competitive Features	Price, customization features	Price, reliability, sales force/tech support quality
Price Sensitivity	High	Medium-High
Perception of client's products	Expensive, sleek, small level of customization, brand inspires quality (although consumers are not sure if computers are of good quality)	Expensive, not appropriate for corporate travel due to less-than-stellar build quality, reliability considered to be low because of "sleek look," inadequate tech support and sales force
Perception of Competitor A's (Dell) products	Affordable, fair quality and reliable, easy to order	Affordable, decent quality, good sales force and excellent tech support
Perception of Competitor B's (HP) products	Expensive, very robust and reliable, decent variety of system configurations, "boxy" looking	Expensive, very robust and reliable, excellent sales force and tech support



Case 45: IBM Meets Lenovo

Additional Information

5. Revenues and Costs

Reveal only the information that the candidate asks for.

As soon as the candidate has information on Average Price, Gross Margins, Market Shares, and Number of Units sold by our client, you should ask him/her to calculate COGS (cost of goods sold) and the number of units sold by Competitors A and B (Dell and HP), and COGS for our client, IBM (for laptops and desktops).

In summary, blue italicized figures should be calculated by the candidate.

	Laptops				Desktops			
	<u>Avg. Price</u>	<u>Gross Margins</u>	<u>COGS</u>	<u>Units</u>	<u>Avg. Price</u>	<u>Gross Margins</u>	<u>COGS</u>	<u>Units</u>
Competitor A (Dell)	\$1,000	9%	<i>\$910</i>	<i>8.0M units/year</i>	\$600	5%	<i>\$570</i>	<i>4.0M units/year</i>
Competitor B (HP)	\$1,500	15%	<i>\$1,275</i>	<i>5.2M units/year</i>	\$900	10%	<i>\$810</i>	<i>2.6M units/year</i>
Our client (IBM)	\$1,900	20%	<i>\$1,520</i>	2.0M units/year	\$1,200	10%	<i>\$1,080</i>	1.0M units/year



Case 45: IBM Meets Lenovo

Recommendation

- Our client, IBM, is losing share in a market where strong competitors such as Dell are thriving through the use of a direct sales model (and high customization level) for consumers, and strong sales force and tech support for corporate customers.
- Not considering Dell, the PC industry has negative average EBITDA margin, evidencing a highly competitive environment.
- Although our client, IBM, has a distinctive brand that is often synonymous to quality and state-of-the-art technology, it has not been possible to realize any price premium without compromising market share.
- IBM's COGS is significantly higher than competitors Dell and HP, possibly due to lower economies of scale. These competitors also specialize in IT and personal computers, and have reached a high level of operational effectiveness (especially Dell), as evidenced by its higher inventory turns and higher EBITDA margins.
- Our client, IBM, is also not aligned with customer needs for both consumers and corporate clients, and would have to significantly improve its tech support and sales force size and effectiveness. Investing in improving its sales force size and effectiveness would lead to high capital investments and would take a long time to ramp up on the learning curve.
- Therefore, our client, IBM, should divest its personal computers division since it seems unlikely that it will be able to compete effectively in that market and the overall market attractiveness is low.
- An outstanding candidate would also consider the implications to IBM's brand, potential companies that would acquire the division, and other alternatives to remain in the PC market, such as alliances, outsourcing production, etc.

Case Book 3

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